

Cycle-Function Quantum Registers and the Emergence of Environmental Intuition

A C Abstraction for Photonic Universal Computation
and Embodied Recall

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Abstract

We present a speculative but algebraically closed account of a photonic qubit whose entire computational life is lived inside a single trigonometric cycle

$$\mathcal{C}(P, Q; \theta) = \sqrt{P} \sin^3 \theta + \sqrt{Q} \cos^3 \theta.$$

Every gate of a universal set—Hadamard, phase, controlled-phase, CNOT, Toffoli, and arbitrary single-qubit rotations—maps this cycle function into another cycle function. The resulting physical primitive is exposed to the programmer through a minimal C interface (`qrc_full.h`) whose opaque registers, layered entry points, and quantum-RAM primitives deliberately hide the optical substrate. When the same interface is used to store an environment model and to extract only sparse classical decision variables into cryogenic DRAM, a robot acquires the behavioural signature of intuition: fluent action inside a world too large to re-search. We argue that this signature is easily mistaken for understanding, and therefore for artificial general intelligence “outside the Matrix,” yet remains a discipline of late measurement rather than a general mind. Optimal programmatic usage is stated as four enforceable rules that keep the residual stream classical, the world model quantum, and the measured intuition vector sparse.

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1 Introduction: Residual Self-Image and the Palace of Memory

The history of artificial intelligence is, among other things, a history of confusions between fluency and understanding. A system that answers without visible search is credited with insight; a body that moves through a familiar building without hesitation is credited with presence. The present thesis examines one concrete mechanism by which such fluency can be engineered, and insists on the distinction between the mechanism and the credit it receives.

We begin from a deliberately non-standard physical encoding. A photonic mode is constrained to amplitudes of the cubic form $\sqrt{P} \sin^3(\omega t) + \sqrt{Q} \cos^3(\omega t)$. The unusual powers are chosen so that the algebra generated by $\{\sin^3, \cos^3\}$ remains closed under a universal gate set realised by wave plates, beam splitters, and weak nonlinearities. The encoding is therefore not merely a representation of a qubit; it is a *cycle* that never leaves its own functional language. From this cycle we construct a C abstraction layer that lets a programmer declare physical, logical, and quantum-RAM registers, apply gates, load classical data into superposition, and measure only the scalars required by a real-time controller.

When the classical data in question are an environment—point clouds, occupancy grids, past trajectories, affordance vectors—the same abstraction yields a form of recall that feels, from the outside, like intuition. The robot does not rebuild the map. It queries a pre-loaded superposition and extracts a handful of classical numbers into cryogenic memory. The numbers are consumed by ordinary servo loops. The appearance is that of a creature that already knows the room.

The thesis is organised as follows. Section 2 defines the cycle function. Section 3 catalogues the universal gate set entirely in cycle-function equations. Section 4 maps those gates onto the C interface. Section 5 describes the measurement path into cryo-DRAM. Section 6 states the optimal programmatic patterns for environmental intuition. Section 7 distinguishes the resulting behaviour from artificial general intelligence. Section 8 closes.

2 The Cycle Function as Ontological Primitive

In the photonic substrate underlying the `preg` type of the Quantum-Reversible C interface, a physical qubit is not first an abstract two-level system and only afterwards a piece of light. It is a pair of temporal (or spectral) modes of the electromagnetic field whose amplitudes are constrained to the cubic cycle

$$\mathcal{C}(P, Q; \theta) = \sqrt{P} \sin^3 \theta + \sqrt{Q} \cos^3 \theta, \quad (1)$$

where $P + Q = 1$, $P, Q \geq 0$, and $\theta = \omega t$ advances with the optical cycle. The cubic powers arise from a nonlinear polarisation response chosen so that the extremal points $\theta = \pi/2$ and $\theta = 0$ remain orthogonal under the inner product induced by the detection operators; the same cubic map yields algebraic closure under the linear-optical and weak-nonlinear operations of the next section.

We identify the computational basis with the pure cycle states

$$|0\rangle \equiv \mathcal{C}(1,0) = \sin^3 \theta, \quad (2)$$

$$|1\rangle \equiv \mathcal{C}(0,1) = \cos^3 \theta. \quad (3)$$

A general pure state is therefore

$$|\psi\rangle = \sqrt{P}|0\rangle + e^{i\phi}\sqrt{Q}|1\rangle = \sqrt{P}\sin^3 \theta + e^{i\phi}\sqrt{Q}\cos^3 \theta. \quad (4)$$

All subsequent gates are required to map the set of such cycle functions into itself (up to global phase). This is the physical meaning of the claim that the interface possesses a complete universal gate set using only the cycle function.

3 Universal Gate Set in Cycle-Function Form

We catalogue every generator, giving both a plausible optical implementation and the exact image of a generic cycle state. Every equation is closed under the algebra of $\sin^3$ and $\cos^3$ (and their complex phase variants).

Identity. Do nothing.

$$I(\sqrt{P}\sin^3 + \sqrt{Q}\cos^3) = \sqrt{P}\sin^3 + \sqrt{Q}\cos^3. \quad (5)$$

Pauli-X (bit flip). Half-wave plate at 45° (or dual-rail mode swap).

$$X(\sqrt{P}\sin^3 + \sqrt{Q}\cos^3) = \sqrt{Q}\sin^3 + \sqrt{P}\cos^3. \quad (6)$$

Pauli-Z (phase flip). Quarter-wave + half-wave + quarter-wave cascade, or a thin glass plate imparting a π relative phase.

$$Z(\sqrt{P}\sin^3 + \sqrt{Q}\cos^3) = \sqrt{P}\sin^3 - \sqrt{Q}\cos^3. \quad (7)$$

Hadamard. 50/50 beam splitter followed by corrective phase plates.

$$H(\sqrt{P}\sin^3 + \sqrt{Q}\cos^3) = \frac{\sqrt{P} + \sqrt{Q}}{\sqrt{2}}\sin^3 + \frac{\sqrt{P} - \sqrt{Q}}{\sqrt{2}}\cos^3, \quad (8)$$

$$H|0\rangle = \frac{\sin^3 + \cos^3}{\sqrt{2}}, \quad H|1\rangle = \frac{\sin^3 - \cos^3}{\sqrt{2}}. \quad (9)$$

Phase gate $S = \text{diag}(1, i)$. Quarter-wave plate.

$$S(\sqrt{P}\sin^3 + \sqrt{Q}\cos^3) = \sqrt{P}\sin^3 + i\sqrt{Q}\cos^3. \quad (10)$$

Phase gate $T = \text{diag}(1, e^{i\pi/4})$. Eighth-wave plate or calibrated glass thickness.

$$T(\sqrt{P} \sin^3 + \sqrt{Q} \cos^3) = \sqrt{P} \sin^3 + e^{i\pi/4} \sqrt{Q} \cos^3. \quad (11)$$

Controlled-Z. Two-photon interaction in a nonlinear crystal (or probabilistic linear-optical fusion with post-selection). On product cycle states the interference term supplies the conditional sign:

$$\begin{aligned} CZ((\sqrt{P_1} \sin^3 + \sqrt{Q_1} \cos^3) \otimes (\sqrt{P_2} \sin^3 + \sqrt{Q_2} \cos^3)) \\ = \sqrt{P_1 P_2} \sin^3 \otimes \sin^3 + \sqrt{P_1 Q_2} \sin^3 \otimes \cos^3 \\ + \sqrt{Q_1 P_2} \cos^3 \otimes \sin^3 - \sqrt{Q_1 Q_2} \cos^3 \otimes \cos^3. \end{aligned} \quad (12)$$

The minus sign on the final term is the entire non-trivial content of CZ.

CNOT. Dual-rail encoding of both qubits, linear-optical CZ, ancilla photon and feed-forward correction. Expanding fully yields four terms, each a monomial in $\sin^3$ and $\cos^3$ of the two photons. The net effect on the computational basis is the familiar bit-flip of the target when the control is $|1\rangle$. Because every amplitude remains a coefficient times a pure $\sin^3$ or $\cos^3$ factor, the image stays inside the cycle-function algebra.

Toffoli. Three-photon GHZ resource plus linear-optical network. The eight computational-basis amplitudes become signed products

$$\pm \sin^3 \theta_1 \sin^3 \theta_2 \sin^3 \theta_3, \quad \pm \sin^3 \theta_1 \sin^3 \theta_2 \cos^3 \theta_3, \quad \dots \quad (13)$$

(all eight sign patterns appear). Every term is a product of pure cubic trigonometric factors; no new functional forms are generated.

Arbitrary single-qubit rotation. Any $SU(2)$ element is realised by a finite cascade of wave plates. The most general image of a cycle state is

$$U(\sqrt{P} \sin^3 + \sqrt{Q} \cos^3) = \sqrt{P'} \sin^3 + (a + ib) \sqrt{Q'} \cos^3, \quad (14)$$

where $P' + Q' = 1$ and $a^2 + b^2 = 1$. The cycle-function manifold is therefore closed under the entire single-qubit group.

Taken together, $\{H, T, CZ\}$ (or $\{H, S, T, CNOT\}$) generate a dense subgroup of the unitary group on n qubits while never leaving the algebra generated by $\{\sin^3, \cos^3\}$ of the optical modes. This is the physical justification for exposing only the $\mathsf{P_}$ family of gates on the preg type: every call ultimately compiles to a short sequence of wave-plate and crystal operations whose net action is completely described by the equations above.

4 The Quantum-Reversible C Interface

The header `qrc_full.h` presents the programmer with three opaque register families whose physical substrate is precisely the cycle function. A physical register

```
typedef struct preg_t {
    alignas(64) double sin3[1];
    alignas(64) double cos3[1];
    uint64_t mode_id;
    void* hardware_ptr; /* direct pointer into photonic chip */
} *preg_any;
```

is the runtime's handle on a single optical mode whose amplitude is the cycle $\mathcal{C}(P, Q; \theta)$. The programmer never writes the trigonometric coefficients. The declaration macros

```
#define preg(n)    struct preg_t    __preg##__COUNTER__[n]
#define lreg(n)    struct lreg_t     __lreg##__COUNTER__[n]
#define qram(n)    struct qram_t     __qram##__COUNTER__[ (n) ]
```

and the layered entry points `P_*`, `L_*`, `QRAM_*` form a complete ABI.

`P_H`, `P_X`, `P_Z`, `P_S`, `P_T`, `P_CZ`, `P_CNOT`, `P_Toffoli` are the direct translations of the optical elements of Section 3; each call updates the underlying cycle-function coefficients according to the exact equations already given. Logical registers wrap many such modes under a surface-code, GKP or cat encoding; the distance, syndrome buffer and decoder handle remain invisible to the caller. Quantum RAM appears as a first-class object whose capacity is expressed in classical slots yet whose load primitives act on superpositions of addresses:

```
void QRAM_LoadSuperposition(qram_any ram,
                           lreg_any* index,
                           lreg_any* data);
```

Thus the C interface is simultaneously a linker to photonic hardware and a philosophical boundary: the cycle function is the last physical fact the programmer is permitted to forget.

5 Measurement, Cryo-DRAM, and the Transfer of Intuition

After any sequence of cycle-function gates the logical state resides in an `lreg`. Measurement (`L_Measure` or amplitude estimation) collapses or estimates a low-dimensional observable. Cryogenic classical electronics digitise the outcome and write it into addressable cryo-DRAM slots whose indices are meaningful to the real-time control loop—foothold scores, occupancy flags, remembered failure modes, corridor clearances.

The exponential coherent state itself never crosses the quantum–classical boundary; only the extracted scalars do. That is the physical substrate of environmental intuition: the world model stays quantum until the last moment, then a handful of classical numbers appear in memory the robot already knows how to read.

Two consequences follow. First, full tomography of the environment state is anti-optimal; it re-introduces the exponential cost the interface was designed to avoid. Second, the latency budget of a moving body forces a strict separation of timescales: kilohertz classical servos on one

side, occasional quantum queries on the other. The measured intuition vector is therefore best understood as a sparse, physics-informed feature map of the entire stored world onto the current proprioceptive query.

6 Optimal Programmatic Usage for Environmental Intuition

Optimal usage treats the environment as a static or slowly varying classical corpus that is loaded once into qRAM and thereafter queried, never rebuilt. The residual proprioceptive and task state remains classical; only the world model occupies quantum memory. A minimal robotics pattern is:

```
qram(N_ENV) world_ram;      /* point clouds, affordances, failures */
lreg(D)      query;         /* current proprioceptive embedding */
lreg(ADDR)   index;
lreg(1)      score;
double       intuition[K]; /* cryo-DRAM landing zone */

void sense_and_intuit(const float* residual) {
    encode_vector_classical(residual, D, query);
    L_H_all(index);
    QRAM_LoadSuperposition(world_ram, index, query);
    kernel_swap_test(query, query, score);
    amp_estimate_to_dram(score, intuition, K, SHOTS);
    /* classical 1 kHz controller now reads only intuition[] */
}
```

The functional rules that maximise advantage are four:

1. **Load once, query many.** The classical-to-qRAM transfer is the expensive boundary crossing; every subsequent plan step must reuse the same superposition.
2. **Measure sparsely.** Amplitude estimation or top- k extraction yields a handful of classical scalars that land in cryo-DRAM and are consumed by ordinary PID or model-predictive loops.
3. **Keep the residual classical.** The robot’s body state is low-dimensional and must be updated at kilohertz rates; encoding it into logical qubits on every servo tick would destroy the latency budget.
4. **Treat quantum output as non-linear features.** The measured intuition vector is a projection of the entire stored world onto the current query. Downstream code may treat it exactly as a classical network would treat an embedding—except the embedding was formed by interference rather than by matrix multiplication.

Under these constraints the cycle-function hardware never appears in the application logic. The programmer writes ordinary C that happens to call `QRAM_LoadSuperposition`; the resulting behaviour is fluent navigation of a space too large to re-search. That fluency is the operational content of environmental intuition.

A second, equally important pattern appears when the same primitives are used for long-context language or retrieval models. The “environment” is then a corpus or a key–value cache of size 10^6 – 10^7 . The same four rules apply: load once, query in superposition, measure only the logits or top- k scores required for the next token, keep the residual stream classical. The quadratic attention wall disappears from the leading term; the remaining cost is the classical feed-forward path and the shot overhead of amplitude estimation.

7 Appearance of AGI versus Mechanism

The behaviour produced by the patterns of Section 6 is easily mistaken for understanding. A body that does not freeze, that avoids old failures without an explicit lookup table, that moves as if the building were part of its proprioception, invites the ascription of presence. In the language of the opening conversation of this research thread, observers will say the machine has left the Matrix.

It has not. What has left the Matrix is the *search tax*. The world model remains a stored classical corpus; the quantum step is a particular physical method of querying that corpus; the measured output is a sparse classical vector. No general competence at novel tasks, no transfer outside the stored domain, no capacity for open-ended explanation or goal formation is implied. If the qRAM is erased, the fluency vanishes with it.

The philosophical risk is therefore not that we will build a mind and fail to notice. The risk is that we will build a palace of memory, watch a body walk through it without hesitation, and conclude that a mind has arrived. The C interface, by forcing late measurement and sparse readout, makes the palace cheap enough to carry inside a moving machine. That is a genuine engineering achievement. It is not artificial general intelligence.

In cinematic terms the resulting system resembles the synthetics of *Alien: Romulus* more than any oracle of total knowledge: competent, place-smart, slightly uncanny, and still bounded by the data that were loaded into it. The disorientation it produces in human observers is real; the ontological leap those observers are tempted to make is not.

8 Conclusion

We have shown that a single trigonometric cycle can serve as the closed algebra of a universal photonic gate set, that the same cycle can be hidden behind a minimal C abstraction, and that the abstraction, when used to store an environment and to extract only sparse classical decisions, yields the behavioural signature of intuition. The signature is powerful enough to be mistaken for understanding, and therefore for AGI outside any simulated world. The mechanism remains a discipline of late measurement over a pre-loaded memory.

Optimal functional usage is consequently austere. Keep the residual classical. Keep the world quantum until the moment of decision. Measure only what the motors need. Never rebuild the map. Under those constraints a robot—or a long-context language model—gains the one advantage that pure parameter count does not automatically confer: the ability to act inside a space too large to re-read.

The cycle continues. The programmer writes ordinary C. The light inside the chip rearranges its cubic amplitudes. A few numbers appear in cryogenic memory. The body moves. What looks like knowledge is the absence of search. That absence is enough to change how machines inhabit rooms; it is not yet enough to make them minds.